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MODULAR FORMS AND MODULAR CURVES

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Abstract. In this paper, we introduce the reader to modular forms and modular curves. The aim of this paper is to develop the theory required to compute the dimension of the space of modular forms. As we go on, we will realise that the dimension formula is a result of a very elegant application of Riemann surfaces, differential forms and the Riemann Roch theorem.

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1. Introduction

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A modular form of weight k is a holomorphic function on the upper half complex plane $\mathbb{H}$. The function transforms in a peculiar way with respect to the action of the group $SL_2(\mathbb{Z})$ on $\mathbb{H}$. We will define the action in section 2 and make this precise in section 3. The classical theory of modular forms is of interest in a variety of areas

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of mathematics. For example, they come up widely in number theory. Wiles proof of the Fermat's last theorem used modular forms extensively. In combinatorics, modular forms are used as tools to prove certain combinatorial identities using theta functions. One of the reason why modular forms are of particular interest is the fact that the space of weight k modular forms is finite dimensional. We can also explicitly calculate the basis for this space.

The aim of this paper is to find the dimension of this space. In order to do so, we introduce modular curves. Suppose Γ is a subgroup of $SL_2(\mathbb{Z})$ of finite index. A modular curve with respect to Γ is the quotient space of orbits under the action of Γ on $\mathbb{H}$. We will see that every modular curve is in fact a Riemann surface. We work with compact modular curves and establish a weight $2k$ modular form as a k-fold differential form on the associated modular curve. The dimension is then calculated in terms of the known data of the modular curve.

This exposition is written as a part of the University of Chicago REU program of 2018. We assume familiarity with complex analysis and basic topology.

2. Action of the modular group on $\mathbb{H}$

Definition 2.1 (The modular group)**.** The modular group is the group of all invertible 2 by 2 matrices with integer entries having determinant 1. We denote it by $SL_2(\mathbb{Z})$.

The modular group acts on the upper half complex plane, $\mathbb{H}$ as follows: For any $\gamma \in SL_2(\mathbb{Z})$ such that $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define:

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}.$$

Lemma 2.2. *The modular group maps the upper half plane to itself.*

Proof. Easy calculations show that $\Im(\gamma\tau) = \frac{(ad-bc)\Im(\tau)}{|c\tau+d|^2}$, where $\Im(\tau)$ denotes the imaginary part of τ. Since $ad - bc = 1, \tau \in \mathbb{H}$ implies that $\Im(\tau) > 0$. We therefore have that $\Im(\gamma\tau) = \frac{\Im(\tau)}{|c\tau+d|^2} > 0$. Thus, $\gamma\tau \in \mathbb{H}$. $\square$

One can easily check the following to confirm that the above defines a group action.

(1) $I(\tau) = \tau$ where I denotes the identity matrix.
(2) $(\gamma\gamma')\tau = \gamma(\gamma'\tau)$, for some $\gamma, \gamma' \in SL_2(\mathbb{Z})$

Lemma 2.3. *The modular group is generated by the matrices:*

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The proof is left as an exercise for the reader.

The first matrix maps τ to $\tau + 1$. In other words, it acts via translation by 1 on τ. The second matrix takes τ to $-1/\tau$. Under this transformation the points inside the unit disc in the upper half plane are mapped to points outside the unit disc and vice versa. These transformations will help us determine the fundamental domain for the action of $SL_2(\mathbb{Z})$ on $\mathbb{H}$ in section 4.1.

p002_L0002 [section-header]
3. Modular forms

p002_L0003 [text]
Definition 3.1 (Modular form of weight k)**.** Let k be an integer. A modular form of weight k, for the group $SL_2(\mathbb{Z})$, is a function satisfying the following:

p002_L0004 [text]
(1) f is holomorphic.

p002_L0005 [formula]
(2) (Modularity condition) For all $\gamma \in SL_2(\mathbb{Z}), \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

p002_L0006 [text]
$$f(\gamma\tau) = (c\tau + d)^k f(\tau) \text{ for all } \tau \in \mathbb{H}.$$

p002_L0007 [list-item]
(3) The function f is holomorphic at ∞. We will make this precise in section 3.1.

p002_L0008 [text]
The first example of a modular form of weight k is the zero function. In fact one can check that for odd weights, the zero function is the only modular form.

p002_L0009 [text]
To check the modularity condition for all the matrices of $SL_2(\mathbb{Z})$ is a very tedious job. However the modularity condition reduces to the following when we check it on the generators.

p002_L0010 [text]
$$f(\tau + 1) = f(\tau) \text{ and } f(-1/\tau) = \tau^k f(\tau) \text{ for all } \tau \in SL_2(\mathbb{Z}). \tag{3.2}$$

p002_L0011 [text]
The next proposition helps us to conclude that checking the modularity condition for all the matrices in $SL_2(\mathbb{Z})$ is equivalent to checking the condition on the generators. We introduce some notation first. Suppose $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $SL_2(\mathbb{Z})$. Let

p002_L0012 [text]
$\jmath(\gamma, \tau) = (c\tau + d)^k$. Denote $(c\tau + d)^{-k} f(\gamma\tau)$ by $f|_\gamma$. The modularity condition for a modular form f is now equivalent to the following statement: $f|_\gamma = f$.

p002_L0013 [text]
Proposition 3.3. *Suppose that f is a modular form of weight k and $\gamma_1, \gamma_2 \in SL_2(\mathbb{Z})$ We have the following:*

p002_L0014 [text]
(1) $\jmath(\gamma_1\gamma_2, \tau) = \jmath(\gamma_1, \gamma_2\tau)\jmath(\gamma_2, \tau)$.

p002_L0015 [text]
(2) $f|_{\gamma_1\gamma_2} = (f|_{\gamma_1})|_{\gamma_2}$.

p002_L0016 [text]
The first part is routine calculation required to prove the next part of the proposition. The second part, however tells us that, if the modularity condition holds for two matrices then it holds for their product as well. Therefore it is now sufficient to look at the conditions in (3.2).

p002_L0017 [text]
It is easy to observe that the set of modular forms of weight k form a vector space over $\mathbb{C}$ with the addition defined by the usual addition of functions.

p002_L0018 [text]
3.1. **The q-expansion of a modular form.** Recall that the map $\tau \mapsto e^{2\pi i\tau}$ is a surjective holomorphic map of the upper half plane to the punctured unit disc. This transformation will help us view modular forms as functions on the unit disc instead of the upper half plane. In this way, we obtain a power series centered at zero. Let $q = e^{2\pi i\tau}$. Since $e^{2\pi i\tau} = e^{2\pi i\tau'}$ if and only if $\tau = \tau' + n$ for some $n \in \mathbb{Z}$, it is therefore well defined to set $\tilde{f}(q) = f(\tau)$. The function f is now defined on the open punctured unit disc via $\tilde{f}$. Note that, as $\tau \to i\infty, q \to 0$. We say that the function f is holomorphic at infinity if $\tilde{f}$ can be analytically extended to the whole unit disc, so that it is analytic at 0. The function therefore can be written as a power series around 0. By abuse of notation we write:

p002_L0019 [formula]
$$\tilde{f}(q) = f(q) = \sum_{n>0} a_n q^n.$$

3.2. **First examples.** For the matrix $-I$, the modularity condition implies that $f(\tau) = (-1)^k f(\tau)$. Therefore, for odd weights the only modular form is the 0 function. For a non-trivial example of a modular form of weight k, consider the double sum:

$$G_k(\tau) = \sum_{\substack{m,n\in\mathbb{Z}\\ m,n\neq 0}} \frac{1}{(m\tau+n)^k}.$$

The series converges absolutely when $k > 2$. We see that when $k > 2$,

$$G_k(\tau+1) = \sum_{\substack{m,n\in\mathbb{Z}\\ m,n\neq 0}} \frac{1}{(m\tau+m+n)^k} = G_k(\tau)$$

The last equality comes from the fact that the series is absolutely convergent, thus we can change the order of the sum over (m,n) to $(m, m+n)$. One can similarly check the second modularity condition. The series $G_k(\tau)$ is known as the Eisenstien series of weight k. A specific example of a modular form of weight 12 is the Δ function given by:

$$\Delta(\tau) = (60G_4(\tau))^3 - 27(140(G_6))^2.$$

We can compute the q-expansion of some Eisenstien series on sage as follows:

```
eisenstein_series_qexp(4,5)
1/240 + q + 9*q^2 + 28*q^3 + 73*q^4 + O(q^5)
eisenstein_series_qexp(6,5)
-1/504 + q + 33*q^2 + 244*q^3 + 1057*q^4 + O(q^5)
```

The above code computes the q-expansion of $G_4(\tau)$ and $G_6(\tau)$ respectively up to the power of five. Sage computes the q-expansion of the Δ function as:

```
delta_qexp(6)
q - 24*q^2 + 252*q^3 - 1472*q^4 + 4830*q^5 + O(q^6)
```

Notice that the constant term a_0 of the q-series for the Δ function is 0. Such forms are called cusp forms.

3.3. **Congruence subgroups of $SL_2(\mathbb{Z})$.** There are some modular forms which satisfy the modularity condition not for the whole group but for a particular subgroup of $SL_2(\mathbb{Z})$. For example, the theta function:

$$\theta(\tau) = \sum_{n=0}^{\infty} r(n,4)e^{2\pi i n\tau}$$

is a modular form of weight 2 with respect to the subgroup generated by the matrices:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$$

The coefficient $r(n,4)$ denotes the number of ways n can be written as a sum of four squares. This is a non trivial result. The above fact and the fact that the space of modular forms is finite dimensional is used to derive an explicit formula for $r(n,4)$. We won't go into the details, but this gives us enough motivation to describe some subgroups of $SL_2(\mathbb{Z})$ of particular interest to us. The subgroup $\Gamma_0(N)$ is described by the following set:

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \;\middle|\; \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \mod N \right\}$$

We also have the subgroup

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \;\middle|\; \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \mod N \right\}$$

The principle congruence subgroup $\Gamma(N)$ is defined as

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \;\middle|\; \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mod N \right\}$$

We will work with a much generalized class of subgroups. These are called the congruence subgroups.

Definition 3.4 (Congruence subgroup)**.** A subgroup Γ of $SL_2(\mathbb{Z})$ is a congruence subgroup of level N if $\Gamma(N) \subset \Gamma$ for some positive integer N.

Fact 3.5. All the subgroups described above are of finite index in $SL_2(\mathbb{Z})$.

3.4. **Meromorphic modular forms.** In this section we define meromorphic modular forms. As the name suggests these are meromorphic functions on the upper half plane satisfying the same properties as holomorphic modular forms. Meromorphicity at ∞ means that the function is now allowed to have a pole at infinity so that f has a Laurant series expansion around 0. One of the first examples of a meromorphic modular form of weight 0 is the j function. It is defined as:

$$j(\tau) = 1728 \frac{(60G_4)^3}{\Delta}(\tau).$$

The first few terms of the q-expansion of the j function are computed on sage as:

```
j_invariant_qexp(3)
```

```
q^-1 + 744 + 196884*q + 21493760*q^2 + O(q^3)
```

Notice that since the j function is of weight 0, i.e., it is $SL_2(\mathbb{Z})$ invariant. Thus it a well defined function on the space $\mathbb{H}/SL_2(\mathbb{Z})$. This space is called a modular curve of level 1. We denote it by $Y(1)$. We will elaborate more on modular curves in the next section.

The set of meromorphic modular forms of weight k with respect to the subgroup Γ also form a vector space. We denote it by $\mathcal{A}_k(\Gamma)$. It contains the subspace of holomorphic modular forms which we denote by $\mathcal{M}_k(\Gamma)$.

4. Modular curves

We know that $SL_2(\mathbb{Z})$ acts on $\mathbb{H}$. Let Γ be a subgroup of $SL_2(\mathbb{Z})$ of finite index.

Definition 4.1 (Modular curves)**.** Define a modular curve $Y(\Gamma)$ to be the quotient of the upper half plane under the action of Γ.

Two points τ and τ' are in the same orbit if and only if there exists some $\gamma \in \Gamma$ such that $\gamma\tau = \tau'$. We give $Y(\Gamma)$ the quotient topology via the map $\Pi : \mathbb{H} \to Y(\Gamma)$. This means that $U \subset Y(\Gamma)$ is open if only if $\Pi^{-1}(U)$ is open in $\mathbb{H}$. Note that Π is an open map.

We next state a very important proposition and its consequences which we will use extensively later. The proof of proposition 4.2 and corollary 4.4 can be found in section 2.1 of [1]

Proposition 4.2. *Let τ_1 and $\tau_2 \in \mathbb{H}$ be given. Then there exist neighborhoods U_1 of τ_1 and U_2 of τ_2 in $\mathbb{H}$ with the property that for any $\gamma \in SL_2(\mathbb{Z})$, if $\gamma(U_1) \cap U_2 \neq \phi$ then $\gamma(\tau_1) = \tau_2$.*

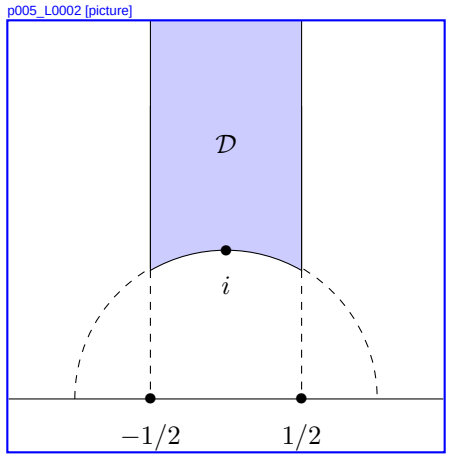


FIGURE 1. The fundamental domain

In simple words, when we take $\tau_1 = \tau_2$ and $U_1 = U_2$, the above proposition guarantees the existence of a neighborhood for every point $\tau \in \mathbb{H}$, in which it is the sole representative of its Γ-orbit. As a direct consequence we have that:

Corollary 4.3. *Let $\tau \in \mathbb{H}$. For any $\gamma \in SL_2(\mathbb{Z})$, there exists a neighborhood U such that $\gamma U \cap U \neq \phi$ implies that γ belongs to Γ_τ, where Γ_τ denotes the stabiliser of τ in $SL_2(\mathbb{Z})$.*

The above proposition is used to show that the modular curves are Hausdorff. It will also help in giving a Riemann structure to the modular curves as we will see in section 4.2.

Corollary 4.4. *The modular curve $Y(\Gamma)$ is Hausdorff where Γ is any congruence subgroup of $SL_2(\mathbb{Z})$.*

4.1. **The fundamental domain.** A fundamental domain for the action of Γ on $\mathbb{H}$ is a region on the upper half plane such that for every point τ in $\mathbb{H}$, the region contains exactly one point which is in the same Γ-orbit as τ. In this section we will find out the fundamental domain for $SL_2(\mathbb{Z})$.

Proposition 4.5. *The fundamental domain for the action of $SL_2(\mathbb{Z})$ is the region $\mathcal{D} = \{\tau \in \mathbb{H} \mid |\Re(\tau)| \leq 1/2, |\tau| \geq 1\}$.*

The region is shown in figure 4.1. We will use standard notations for some of the matrices which we will encounter a lot. Let

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}; \; T = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \tag{4.6}$$

Proof. We first show that for every $\tau \in \mathbb{Z}$ there exists $\tau' \in \mathcal{D}$ such that $\tau = \gamma\tau'$ for some $\gamma \in SL_2(\mathbb{Z})$. This will show that any point in $\mathbb{H}$ is $SL_2(\mathbb{Z})$ equivalent to a point in $\mathcal{D}$. Choose any $\tau \in \mathbb{H}$ and apply the matrix T^n for some $n \in \mathbb{Z}$ to translate τ by n to the vertical strip $\{\tau \mid |\Re(\tau)| \leq 1/2\}$. If τ translates into $\mathcal{D}$, then we are done. If not, then $|\tau| < 1$. Now notice that $\Im(-1/\tau) = \Im(\tau)/|\tau|$. Thus when $|\tau| < 1, \Im(-1/\tau) > \Im(\tau)$. Therefore, the next step would be to repeatedly apply the matrix S until $\Im(\tau) > 1$. We need to make sure the above process stops at a finite

number of steps. To see this, notice that for any τ there are only finite number of integer pairs (c, d) such that $|c\tau + d| < 1$. Since $\Im(\gamma\tau) = \Im(\tau)/|c\tau + d|^2$, thus there are only finitely many transformations such that $\Im(\gamma\tau) > \Im(\tau)$. Therefore the above process terminates at some point. □

However, notice that the line $\Re(\tau) = 1/2$ is identified with the line $\Re(\tau) = -1/2$. The two halves of the boundary arc $\tau = 1$ are also identified via $\tau \mapsto -1/\tau$. These are the only two identifications. This is a bit tedious to check. See section 2.3 of [1] for the proof.

4.2. **Realising a modular curve as a Riemann surface.** A Riemann surface is a Hausdorff, second countable topological space, which locally "looks like" the complex plane. In order to give modular curves a Riemann structure, we need to make sure that every point on the modular curve has a neighborhood homeomorphic to an open set in the complex plane. These are called local charts which define a local coordinate structure on the Riemann surface. More precisely, a Riemann surface X consists of the following:

(1) A collection $(U_i, V_i, \phi_i)_{i\in I}$ for some indexing set I, where $U_i \in \mathbb{C}$ are open, V_i are open in X and $\phi_i : V_i \to U_i$ are homeomorphisms. We call V_i charts and the set $\{V_i\}_{i\in I}$, an atlas on X.

(2) If for any two charts $V_i \cap V_j \neq \phi$, then the map $\phi_j^{-1} \circ \phi_i : \phi_i(U_i \cap U_j) \to \phi_j(U_i \cap U_j)$ is holomorphic.

Condition 2 helps us to smoothly go from one chart to another via the intersection. The simplest example of a Riemann surface is the complex sphere. The torus is also a Riemann surface. In fact, any g holed surface is an example of a Riemann surface. For a quick introduction to Riemann surfaces refer to [5]. We will later see that when we compactify the modular curve $Y(1)$, it is topologically a sphere.

It is simplest to define a coordinate chart around the points whose stabiliser consist only of $\pm I$. We define local charts around such points as follows: Take a neighborhood U of τ which satisfies the condition in corollary 4.3. Suppose V is its image in $Y(\Gamma)$. Define $\phi = \Pi|_U$.

Claim 4.7. The map $\phi : U \to V$ is a homeomorphism.

Proof. The map ϕ is clearly a surjection. Let τ_1 and τ_2 be two points in U such that $\Pi(\tau_1) = \Pi(\tau_2)$. Then $\tau_1 = \gamma\tau_2$ for some $\gamma \in \Gamma$. We know that $\gamma' U \cap U = \phi$ for all $\gamma' \in \Gamma/\{\pm I\}$. Corollary 4.3 implies that γ is I or $-I$. Thus $\tau_1 = \tau_2$. The function ϕ is therefore an injection as well. Since Π is an open map, we have that ϕ is a homeomorphism from U to V. □

We have a slight problem when the stabiliser of a point τ in the upper half plane is a non trivial subgroup of the modular group. For example, consider the point $i \in \mathbb{H}$ and the transformation S given in (4.6). One can easily see that $Si = i$ i.e., $S \in \Gamma_i$. If we take any neighborhood around i, S acts as a rotation by π around i. To see this, assume that U is any neighborhood around i. Notice that the transformation $\delta_i = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$ takes i to 0. Thus $\delta_i(U)$ is a neighborhood around 0. Also, δ_i is a homeomorphism from U to $\delta_i(U)$. The transformation S fixes i if and only if $\delta_i S \delta_i^{-1}$ fixes 0. Therefore S acts on the point τ in U in the same way as $\delta_i S \delta_i^{-1}$ acts on the point $\delta_i\tau$ in $\delta_i(U)$. We can easily calculate that $\delta_i S \delta_i^{-1}$ maps z in $\delta_i(U)$ to $-z$ i.e., it acts as a rotation by π about 0.

The above argument shows that we will always have Γ-equivalent points in any neighborhood around i and we cannot use the previous argument to define a local coordinate around i. In general, the points with a non-trivial stabiliser are called elliptic points.

Definition 4.8 (Elliptic point)**.** We say $\tau \in \mathbb{H}$ is an elliptic point if $\Gamma_\tau/\{\pm I\}$ is not equal to $\{I\}$.

4.3. **More on elliptic points.** In this section we elaborate on elliptic points. We will give some idea on how to calculate elliptic points for the modular curve $Y(1)$.

4.3.1. *Finding elliptic points for* $Y(1)$. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a non-trivial matrix in $SL_2(\mathbb{Z})$ such that $\gamma\tau = \tau$ for some $\tau \in \mathbb{H}$. After solving the equation $a\tau + b/c\tau + d = \tau$, we get the quadratic equation $c\tau^2 + (d-a)\tau - b = 0$. If $c = 0$, then $a = d = \pm 1$ gives us that $\tau + b = \tau$. This implies that $b = 0$ and $\gamma = \pm I$. Since we assumed γ to be nontrivial, we have two distinct solutions to the above quadratic equation given by:

$$\tau = \frac{-(d-a) \pm \sqrt{(d-a)^2 + 4bc}}{2c}$$

The term $(d-a)^2 + 4bc < 0$ as $\tau \in \mathbb{H}$. On simplifying we get that $|a+d| < 2$. We therefore have that $a + d = \pm 1$ or 0.

Next, notice that the characteristic polynomial of γ is $x^2 - (a+d)x + 1$. So the only possibilities for the characteristic polynomial are x^2+1, x^2-x+1 or x^2+x+1. The polynomial x^2+1 is a factor of x^4-1 while x^2-x+1 and x^2+x+1 are factors of x^6-1. By Caley Hamilton theorem, γ satisfies its own characteristic polynomial. Thus $\gamma^4 = I$ or $\gamma^6 = I$. Therefore the possible orders for γ are $1, 2, 3, 4$ or 6. One can check that $\gamma = \pm I$ when γ has order 1 or 2.

The next theorem characterizes the matrices of order $3, 4$ or 6 in $SL_2(\mathbb{Z})$. The proof of the theorem can be found in section 2.3 of [1].

Theorem 4.9. *Let $\gamma \in SL_2(\mathbb{Z})$. Denote the following matrices as:*

$$R = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}; S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}; W = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$

(1) if γ has order 3, then γ is conjugate to $R^{\pm 1}$.
(2) if γ has order 4, then γ is conjugate to $S^{\pm 1}$.
(3) if γ has order 6, then γ is conjugate to $W^{\pm 1}$.

An immediate consequence of the above theorem is the following corollary:

Corollary 4.10. *The elliptic points for $SL_2(\mathbb{Z})$ are the points $SL_2(\mathbb{Z})i$ and $SL_2(\mathbb{Z})\mu_3$ where $\mu_3 = e^{2\pi i/3}$. The modular curve $Y(1)$ therefore has just two elliptic points.*

Proof. The matrices R and W fix μ_3, while S fixes i. Recall that any matrix that fixes some point $\tau \in \mathbb{H}$ is conjugate to R, S or W in $SL_2(\mathbb{Z})$. Suppose δ is a transformation in $SL_2(\mathbb{Z})$ such that it is conjugate to S, i.e., $\delta = \gamma S \gamma^{-1}$ for some $\gamma \in SL_2(\mathbb{Z})$. Then δ fixes γi. Similarly, we argue for W and R and conclude that the elliptic points for the modular group are $SL_2(\mathbb{Z})i$ and $SL_2(\mathbb{Z})\mu_3$.

From the discussion in section 4.1 we know that the orbits of i and μ_3 are not equal in $Y(1)$. Therefore $Y(1)$ has two elliptic points. $\square$

p008_L0001 [text]

It is easy to check the following corollary which explicitly calculates the stabilisers of the elliptic points i and μ_3.

p008_L0002 [text]

Corollary 4.11. *The stabiliser of i in $SL_2(\mathbb{Z})/\{\pm I\}$ is the cyclic subgroup of order 2, generated by the matrix S. The stabiliser of μ_3 in $SL_2(\mathbb{Z})/\{\pm I\}$ is the cyclic subgroup of order 3, generated by the matrix W.*

p008_L0003 [text]

Notice that this implies that the stabiliser $SL_2(\mathbb{Z})_\tau$ of any elliptic point τ is a finite cyclic group of order 2 or 3.

p008_L0004 [text]

Suppose Γ is any congruence subgroup of $SL_2(\mathbb{Z})$. The elliptic points of Γ are a subset of the points $SL_2(\mathbb{Z})i$ and $SL_2(\mathbb{Z})\mu_3$. Since Γ has a finite index in $SL_2(\mathbb{Z})$, we can conclude that the number of elliptic points in $Y(\Gamma)$ is finite. Otherwise they would be in infinitely many Γ-orbits in $SL_(\mathbb{Z})$ which is not possible. For any elliptic point τ, the stabiliser Γ_τ is a subgroup of $SL_2(\mathbb{Z})_\tau$. Therefore Γ_τ is finite cyclic. From the above discussion we arrive at a very important corollary:

p008_L0005 [text]

Corollary 4.12. *Let Γ be a congruence subgroup of $SL_2(\mathbb{Z})$. For each elliptic point τ of Γ, its stabiliser Γ_τ is finite cyclic.*

p008_L0006 [text]

Thus to each τ, we associate a positive integer e_τ such that e_τ counts the number of τ fixing transformations. Keeping in mind that $-I$ acts trivially, $e_\tau = |\Gamma_\tau|$ if $-I \notin \Gamma$. Otherwise, $e_\tau = |\Gamma_\tau|/2$. Notice that $e_\tau > 1$ if and only if τ is elliptic. The integer e_τ is called the period of τ.

p008_L0007 [text]

Remark 4.13. If τ is an elliptic point, the only possibilities for the value of e_τ are 2 and 3. So the elliptic points for any modular curve are either of period two or three.

p008_L0008 [text]

Let $\Gamma_0(N)$ be the subgroup as described in section 3.3. Sage computes the number of elliptic points of period 2 of $Y(\Gamma_0(N))$ for values of N ranging from 1 to 20 as:

p008_L0009 [text]

```
[Gamma0(n).nu2() for n in [1..20]]
[1, 1, 0, 0, 2, 0, 0, 0, 0, 2, 0, 0, 2, 0, 0, 0, 2, 0, 0, 0]
```

The number of elliptic points of period 3 for the same subgroups is computed as:

```
[Gamma0(n).nu3() for n in [1..20]]
[1, 0, 1, 0, 0, 0, 2, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 2, 0]
```

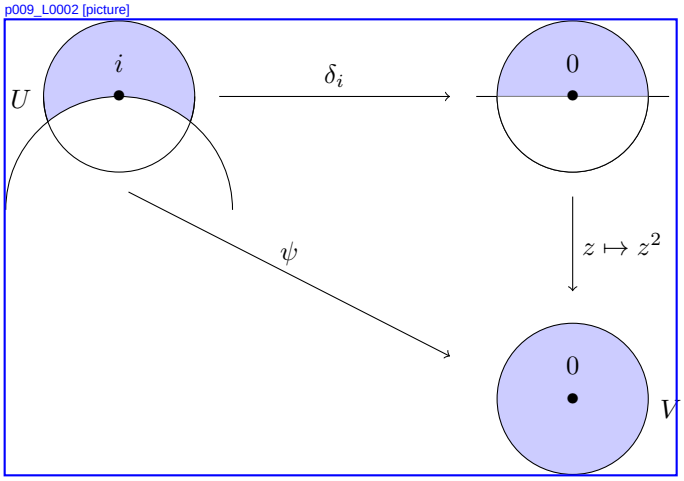
Similarly, for the modular curve $Y(\Gamma_1(N))$ we have:

```
[Gamma1(n).nu2() for n in [1..20]]
[1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[Gamma1(n).nu3()for n in [1..20]]
[1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

p008_L0010 [text]

4.4. **Defining coordinate charts for elliptic points.** The goal of this section is to define coordinate charts around elliptic points. We generalize the argument given for the point i at the end of section 4.2 to define a coordinate chart around elliptic points. The idea is as follows: Suppose τ is an elliptic point. We will take a neighborhood U of τ such that it contains no other elliptic points (such a neighborhood exists, as we will see later). We space out the Γ-equivalent points by a fixed angle $2\pi/e_\tau$. We then wrap the sector of angle $2\pi/e_\tau$ of U around a disc V. This will induce a homeomorphism from $\Pi(U) \to V$. Figure 4.4 shows the coordinate chart around the elliptic point i.

p009_L0002 [picture]



p009_L0003 [caption]
FIGURE 2. Coordinate chart around i

p009_L0004 [text]
Claim 4.14. For any $\tau \in \mathbb{H}$, the period e_τ is well defined on the modular curve $Y(\Gamma)$.

p009_L0005 [text]
Proof. We need to show that for any $\tau \in \mathbb{H}$, the period of $\delta\tau$ is same as the period of τ for all $\delta \in \Gamma$. Suppose $\gamma \in \Gamma_\tau$. We then have that $\gamma\tau = \tau$ if and only if $\delta\gamma\delta^{-1}(\delta\tau) = \delta\tau$. Thus $\delta\Gamma_\tau\delta^{-1} = \Gamma_{\delta\tau}$. Notice that $|\delta\Gamma_\tau\delta^{-1}| = |\Gamma_\tau|$. Therefore, $|\Gamma_\tau| = |\Gamma_{\delta\tau}|$ showing that the periods of τ and $\delta\tau$ are same. $\square$

p009_L0006 [text]
Suppose τ is any elliptic point. Take a neighborhood U of τ given in corollary 4.3. Consider the transformation $\delta_\tau = \begin{pmatrix} 1 & -\tau \\ 1 & -\bar{\tau} \end{pmatrix}$. This takes $\tau \to 0$ and $\bar{\tau} \to \infty$ acting as the same way as the modular group. Notice that $\delta_\tau(U)$ is a neighborhood around the origin. The stabiliser of $\delta_\tau\tau$ is $\delta_\tau\Gamma_\tau\delta_\tau^{-1}$. We will use the following observation: Γ_τ acts on the point z in U in the same way as $\delta_\tau\Gamma_\tau\delta_\tau^{-1}$ acts on δz.

The transformation $\delta_\tau\Gamma_\tau\delta_\tau^{-1}$ fixes $\delta\tau = 0$ and $\delta\bar{\tau} = \infty$. Therefore, $\delta_\tau\Gamma_\tau\delta_\tau^{-1}$ is a transformation of the form $z \mapsto cz$ for some complex number c. By corollary 4.12, we know that $\delta_\tau\Gamma_\tau\delta_\tau^{-1}$ is a finite cyclic group of order e_τ. It is therefore the group of rotations by $2\pi/e_\tau$ about 0. Thus Γ_τ corresponds to a rotation of $2\pi/e_\tau$ about τ. When we transform U via the map δ_τ, the Γ-equivalent points are separated by a fixed angle.

p009_L0007 [text]
We next wrap the sector around a disc V via the map $z \mapsto z^{e_\tau}$. Call this map ρ. Define $\psi = \rho \circ \delta$. Since it is clear that δ_τ is defined for a particular point τ, for simplicity, we drop the subscript. Note that by the Open mapping theorem ψ is an open map. Our final claim for this section is the following:

p009_L0008 [text]
Claim 4.15. The projection $\Pi : U \to \Pi(U)$ and the map $\psi : U \to V$ identify the same points.

p009_L0009 [text]
Proof. We need to show that for any two points τ_1 and τ_2 in U, $\Pi(\tau_1) = \Pi(\tau_2)$ if and only if $\psi(\tau_1) = \psi(\tau_2)$. Notice that $\Pi(\tau_1) = \Pi(\tau_2)$ if and only if $\tau_1 \in \Gamma\tau_2$. Since U is a neighborhood as in corollary 4.3, we see that $\tau_1 \in \Gamma_\tau\tau_2$. This happens if and

p010_L0001 [text]
only if $\delta\tau_1 \in \delta\Gamma_\tau\delta^{-1}(\delta\tau_2)$ if and only if $\delta\tau_1 = e^{2\pi i d/e_\tau}\delta\tau_2$ for some positive integer d. This implies that $\Pi(\tau_1) = \Pi(\tau_2)$ if and only if $(\delta\tau_1)^{e_\tau} = (\delta\tau_2)^{e_\tau}$ if and only if $\psi(\tau_1) = \psi(\tau_2)$. □

p010_L0002 [text]
The above claim helps us to induce an injection $\phi : \Pi(U) \to V$ such that the diagram below commutes. In fact ϕ is an open map and a surjection as well since Π and ψ are so. Therefore, ϕ is a homeomorphism from $\Pi(U)$ to V.

p010_L0003 [formula]
$$\begin{tikzcd} & U \arrow[dl, "\Pi"'] \arrow[dr, "\psi"] & \\ \Pi(U) \arrow[rr, "\phi"'] & & V \end{tikzcd}$$

p010_L0004 [text]
To see the proof of the fact that the map defined above agrees on the overlaps refer to section 2.2 of [1].

p010_L0005 [text]
One point to keep in mind is that for elliptic points, the local homeomorphism looks like $z \mapsto z^e$ for some positive integer e.

p010_L0006 [text]
4.5. **Cusps.** In order to make the modular curve compact, we add the points $\{\infty\} \cup \mathbb{Q}$ to the the upper half plane. Call $\mathbb{H} \cup \{\infty\} \cup \mathbb{Q}$ the extended upper half plane and denote it by $\hat{\mathbb{H}}$. Action of $SL_2(\mathbb{Z})$ on rationals is defined as:

p010_L0007 [formula]
$$\begin{pmatrix} a & b \\ c & b \end{pmatrix} \left(\frac{m}{n}\right) = \frac{am + bn}{cm + dn}$$

p010_L0008 [text]
The point ∞ goes to a/c. Also notice that $-c/d$ goes to ∞. The point infinity is only identified with rationals up to the action of $SL_2(\mathbb{Z})$ or any of its subgroups. This justifies adding rationals to the upper half plane.

p010_L0009 [text]
Define the compact modular curve $X(\Gamma)$ to be $\hat{\mathbb{H}}/\Gamma$. The points $\{\infty\} \cup \mathbb{Q}$ up to Γ-equivalence are called cusps of $X(\Gamma)$.

p010_L0010 [text]
Proposition 4.16. *The modular curve $X(1)$ has one cusp.*

p010_L0011 [text]
Proof. Let $s = a/c$ be a cusp such that $\gcd(a, c) = 1$. Then there exist integers b, d such that $ad - bc = 1$. Define:

p010_L0012 [formula]
$$\gamma_s = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

p010_L0013 [text]
Then $\gamma_s(\infty) = s$ and $\gamma_s \in SL_2(\mathbb{Z})$. Thus every cusp is $SL_2(\mathbb{Z})$-equivalent to the point ∞ and we have just one cusp in $X(1)$. □

p010_L0014 [text]
Since any congruence subgroup Γ has a finite index in $SL_2(\mathbb{Z})$, a similar argument as done for elliptic points shows that $X(\Gamma)$ has finite number of cusps. We give some examples of finding the representatives of cusps for some of the modular curves using sage. For $X(\Gamma_0(4))$ the representatives are:

```
Gamma0(4).cusps()
[0, 1/2, Infinity]
```

Similarly for $X(\Gamma_1(7))$ we have:

```
Gamma1(7).cusps()
[0, 2/7, 1/3, 3/7, 1/2, Infinity]
```

We can also calculate the number of cusps. For example the number of cusps of $X(\Gamma_0(N))$ for N ranging from 1 to 20 are:

```
[Gamma0(n).ncusps() for n in [1..20]]
[1, 2, 2, 3, 2, 4, 2, 4, 4, 4, 2, 6, 2, 4, 4, 6, 2, 8, 2, 6]
```

We similarly compute the cusps of $X(\Gamma_1(N))$ and $X(\Gamma(N))$ for N ranging from 1 to 15.

```
[Gamma1(n).ncusps() for n in [1..15]]
[1, 2, 2, 3, 4, 4, 6, 6, 8, 8, 10, 10, 12, 12, 16]
```

```
[Gamma(n).ncusps() for n in [1..15]]
[1, 3, 4, 6, 12, 12, 24, 24, 36, 36, 60, 48, 84, 72, 96]
```

4.6. **Defining a topology on the extended upper half plane.** In order to define coordinate charts around the cusps, we first need to define open sets around them. For any arbitrary $M > 0$, define $\mathcal{N}_M = \{\tau \in \mathbb{H} \mid \Im(\tau) > M\}$. Take $\mathcal{N}_M \cup \{\infty\}$ to be the open set around the point infinity. Around the rationals, we take all the sets of the form $\alpha(\mathcal{N}_M \cup \{\infty\})$ where $\alpha \in SL_2(\mathbb{Z})$. The transformation α takes the upper half plane plane at $\Im(\tau) = M$ to a disc tangent at a rational number s where s is such that $\alpha(\infty) = s$. We next induce quotient topology to the modular curve $X(\Gamma)$ via the natural map $\Pi : \hat{\mathbb{H}} \to X(\Gamma)$.

Proposition 4.17. *The modular curve $X(\Gamma)$ is Hausdorff, connected and compact.*

The proof of the proposition can be found in section 2.4 of [1]. We next define charts around cusps to make $X(\Gamma)$ into a compact Riemann surface.

4.7. **Defining coordinate charts for cusps.** At the cusps, infinitely many sectors come together. See figure 4.7. Let s be a cusp. In order to define a local coordinate at s, we first take the neighborhood $\mathcal{N}_2 \cup \{\infty\}$ of ∞. Let δ be a transformation such that $\delta s = \infty$. Let U_s be the neighborhood $\delta^{-1}(\mathcal{N}_2 \cup \{\infty\})$. In this case δ straightens the neighborhood by making the identified sectors differ by a "translation" by a positive integer h. We then act the transformation $\rho = e^{2\pi i\tau/h}$ which "wraps" the neighborhood of infinity into a disc with centre 0. To see this precisely, we first introduce the notion of the width of a cusp.

Notice that the stabiliser of the point ∞ is the subgroup:

$$SL_2(\mathbb{Z})_\infty = \left\{ \pm \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \middle| \, m \in \mathbb{Z} \right\}$$

These are the exactly the translations in $SL_2(\mathbb{Z})$ as they take $\tau \mapsto \tau + m$.

Remark 4.18. The reader might be wondering why we specifically chose the neighborhood $\mathcal{N}_2$ of infinity. It is because $\mathcal{N}_2$ has no elliptic points. This fact helps us prove that any if τ_1 and τ_2 in $\mathcal{N}_2$ are identified by a transformation γ, then γ is a translation. It is also used to prove that the transition map is holomorphic. See section 2.4 of [1] for the explicit proof.

Definition 4.19 (Width of a cusp)**.** Suppose s is a cusp for the modular curve $X(\Gamma)$. Let δ be a transformation taking s to ∞. We define:

$$h_s = |SL_2(\mathbb{Z})_\infty / (\delta\{\pm\}\Gamma\delta^{-1})_\infty|.$$

The width h_s has the following properties. Refer to [7] for the proof of these properties.

Proposition 4.20. *The width h_s is characterized by the following properties:*

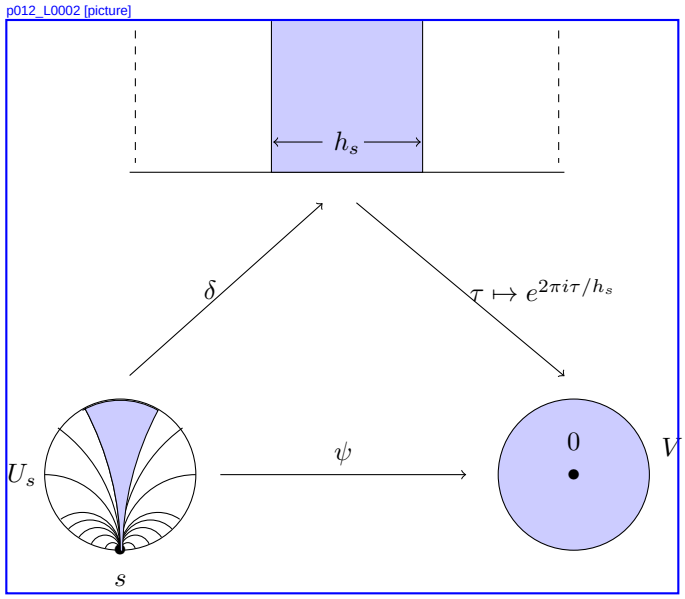


FIGURE 3. Coordinate chart around a cusp s.

(1) It is finite.
(2) It is independent of the transformation δ.
(3) It is well defined on $X(\Gamma)$.

Define $\psi = \rho \circ \delta$ where ρ and δ are transformations described in the beginning of this section. We next claim that:

Claim 4.21. The maps ψ and Π carry out the same identification in the neighborhood U_s of s.

The argument given below will also help in realizing the geometric significance of h_s.

Proof. Suppose τ_1 and τ_2 are two points in U_s, then $\Pi(\tau_1) = \Pi(\tau_2)$ if and only of $\tau_1 = \gamma\tau_2$ for some $\gamma \in \Gamma$. This happens if and only if $\delta\tau_1 = (\delta\gamma\delta^{-1})(\delta\tau_2)$. Since $\delta\tau_1$ and $\delta\tau_2$ both lie in $\mathcal{N}_2$, the transformation $\delta\gamma\delta^{-1}$ is a translation. We can now conclude that $\delta\gamma\delta^{-1} \in \delta\Gamma\delta^{-1} \cap SL_2(\mathbb{Z})_\infty = (\delta\Gamma\delta^{-1})_\infty$. This is exactly the subgroup generated by $\pm\begin{pmatrix} 1 & h_s \\ 0 & 1 \end{pmatrix}$ by the definition of the width. This implies that $\Pi(\tau_1) = \Pi(\tau_2)$ if and only if $\delta(\tau_1) = \delta(\tau_2) + mh_s$ for some $m \in \mathbb{Z}$, if and only if $\psi(\tau_1) = \psi(\tau_2)$. $\square$

As discussed at the beginning of the section, it is clear now that δ is spacing out the Γ-equivalent sectors by a translation by h_s. The claim 4.21 helps us to induce an injection ψ as we did in the case of elliptic points, from $\Pi(U_s)$ to V. Arguing similarly as before, ψ is a local homeomorphism from $\Pi(U_s)$ to V.

4.8. **Genus.** We have finally established a modular curve as a compact Riemann surface. We noted in section 4.2 that a g holed surface is an example of a Riemann surface. We in fact have that:

p013_L0002 [text]
Fact 4.22. Every compact Riemann surface looks like a g holed surface for some positive integer g. The number g is called the genus of the surface.

p013_L0003 [text]
After doing the identifications on the fundamental domain, we get the modular curve $Y(1)$ which is topologically equivalent to a punctured sphere. When we add the cusp to make it compact, we get a sphere. Therefore $X(1)$ is a surface of genus 0. In fact the j function defined in section 3.4 is a well defined homeomorphism from $X(\Gamma)$ to the complex sphere. One can refer to section 3.2 of [1] for the details. To see some more examples we calculate the genus of $X(\Gamma_0(N))$ for values of N ranging from 1 to 20.

p013_L0004 [text]
```
[Gamma0(n).genus() for n in [1..20]]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1]
```
Sage similarly calculates the genus of $X(\Gamma_1(N))$ and $X(\Gamma(N))$ respectively for N ranging from 1 to 20.

p013_L0005 [text]
```
[Gamma1(n).genus() for n in [1..20]]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 2, 1, 1, 2, 5, 2, 7, 3]
```

p013_L0006 [text]
```
[Gamma(n).genus() for n in [1..20]]
[0, 0, 0, 0, 0, 1, 3, 5, 10, 13, 26, 25, 50, 49, 73, 81, 133, 109,
 196, 169]
```

p013_L0007 [section-header]
5. Modular forms as k-fold differentials

p013_L0008 [section-header]
5.1. A formal introduction to differential forms.

p013_L0009 [text]
Definition 5.1 (Meromorphic differential form)**.** A Meromorphic differential form on an open subset U of $\mathbb{C}$ is an expression of the form $f(z)dz$ where f is a meromorphic function on U.

p013_L0010 [text]
Suppose $\phi : U \to U'$ is a mapping from U to another open set $U' \subset \mathbb{C}$. Let $z' = \phi(z)$. Let $\omega = f(z')dz'$ be a differential form on U'. Then $\phi^*(\omega)$ is a differential form defined as $f(\phi(z))\omega'(z)dz$ on U. The differential from $\phi^*(\omega)$ is called the pullback of ω.

p013_L0011 [text]
Since any Riemann surface has local coordinates (U_i, ϕ_i), we define local differential forms ω_i on each homeomorphic image of U_i in $\mathbb{C}$ which agree on the overlaps.

p013_L0012 [text]
Consider a differential form $\omega = f(z)dz$ on $\mathbb{H}$ where $f(z)$ is a meromorphic function. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$. When we pull back the differential ω via the action of γ we get that:

p013_L0013 [section-header]
$$\gamma^*(\omega) = f(\gamma z)d(\gamma z) = f(\gamma z)(cz+d)^{-2}$$

p013_L0014 [text]
So $\gamma^*(\omega) = \omega$ if and only if $f(\gamma z) = (cz+d)^2 f(z)$ i.e., f is a meromorphic modular form of weight 2. We in fact have a one-one correspondence between meromorphic modular forms of weight 2 and meromorphic differential forms on $\mathbb{H}$ invariant under the action of $SL_2(\mathbb{Z})$, which are precisely the meromorphic differentials on $X(1)$. We generalize this to weight k modular forms with respect to the subgroup Γ by formally defining k-fold differential forms.

p013_L0015 [text]
Definition 5.2 (k-fold differential forms)**.** Define a k-fold differential form by the symbol $f(z)dz^k$ on an open set $U \subset \mathbb{C}$ where f is a meromorphic function on U.

Denote the space of k-fold meromorphic differential forms on U by $\Omega^k(U)$.

Suppose X is a Riemann surface. Let (U_i, V_i, ϕ_i) be its coordinate charts. A k-fold meromorphic differential ω on X is a collection of local meromorphic differentials $(\omega_j)_{j\in J}$ that are compatible on the overlaps. Each $\omega_i = f_i(z_i)dz_i$ where z_i is the local coordinate on U_i.

We denote the set of k-fold differential forms on the modular curve $X(\Gamma)$ by $\Omega^k(X(\Gamma))$. We next state a very important theorem which helps us to associate weight $2k$ modular forms with k-fold differential forms.

Theorem 5.3. *Let $k \in \mathbb{Z}$. Let Γ be a congruence subgroup of $SL_2(\mathbb{Z})$. Let $\Pi : \hat{\mathbb{H}} \to X(\Gamma)$ be the projection map. Then the map:*

$$\mathcal{A}_{2k}(\Gamma) \to \Omega^k(X(\Gamma))$$

is an isomorphism of complex vector spaces. Under this isomorphism any meromorphic modular form $f \mapsto (\omega_j)_{j\in J}$ where $(\omega_j)_{j\in J}$ pulls back to the differential form $f(\tau)d\tau^k \in \Omega^k(\hat{\mathbb{H}})$ via Π^*.

Notice that $\Pi^* : \Omega^k(X(\Gamma)) \to \Omega^k(\hat{\mathbb{H}})$ pulls back a differential form on $X(\Gamma)$ to a differential form on $\hat{\mathbb{H}}$. Since the pullback comes from $X(\Gamma)$, it is Γ-invariant i.e., $\gamma^*(f(\tau))(d\tau)^k = f(\tau)(d\tau)^k$. Therefore f satisfies the modularity condition for the integer $2k$. We associate the modular form f to the differential form ω via Π^*. The above theorem says that we have a one to one correspondence between the set of meromorphic modular forms of weight $2k$ and the set of k-fold differential forms on the modular curve $X(\Gamma)$. We will use this isomorphism of vector spaces to find the dimension of the space $\mathcal{M}_{2k}(\Gamma)$. We next state an important fact about the structure of the space $\mathcal{A}_{2k}(\Gamma)$. This will help us to characterize the corresponding space of k-fold differential forms.

Fact 5.4. Let f be any non zero element in $\mathcal{A}_{2k}(\Gamma)$. Let $C(X(\Gamma))$ denote the field of meromorphic functions on $X(\Gamma)$. Then $\mathcal{A}_{2k}(\Gamma) = C(X(\Gamma))f$.

From the above fact and theorem 5.3 we conclude that $\Omega^k(X(\Gamma)) = C(X(\Gamma))\omega_0$ where ω_0 is some non zero k-fold differential form on $X(\Gamma)$.

6. Divisors and the Riemann Roch theorem

We next introduce the Riemann Roch theorem. The aim of this section is to formally develop the necessary definitions in order to make sense of the Riemann Roch theorem. We first state some facts regarding the zeroes and the poles of functions on a compact Riemann surface X.

Fact 6.1. Any holomorphic function on X is constant.

Fact 6.2. A meromorphic function f on X has finitely many poles and zeroes.

Fact 6.3. Any meromorphic function f on X has the same number of poles as the zeroes counting with multiplicities.

Definition 6.4 (Divisor)**.** Let X be a compact Riemann surface. A divisor D on X is a finite sum over the points of X formally written as

$$\sum_i n_i \cdot P_i$$

where $n_i \in \mathbb{Z}$ and P_i are points of X.

The divisor $D = \sum n_i \cdot P_i$ is called positive if $n_i \geq 0$ for all i. In such a case we write $D \geq 0$.

Definition 6.5 (Order of f at a point)**.** Let f be a meromorphic function on X. For any point $P \in X$, order of f at P is the order of the pole or zero at P. It is denoted by $\mathrm{ord}_P(f)$. It is positive if P is a zero of f, negative if P is a pole or zero otherwise.

Definition 6.6 (Divisor of f)**.** Let f be a non zero meromorphic function on X. The divisor of f is the formal sum:

$$\sum_P \mathrm{ord}_P(f) \cdot P.$$

We denote it by $\mathrm{div}(f)$.

The sum makes sense because of fact 6.2.

Remark 6.7. We will see in section 7 that the divisor associated to a holomorphic modular form is not integral i.e., its coefficients are not integral. In such a case we take the greatest integer of the coefficients. This makes sense because $\mathrm{div}(f) \geq 0$ if and only if $\lfloor \mathrm{div}(f) \rfloor \geq 0$.

Definition 6.8 (Degree of a divisor)**.** For any divisor $D = \sum_i n_i \cdot P_i$, the degree of D is the sum $\sum_i n_i$.

Note that by fact 6.3, the degree of a meromorphic function is 0.

We next associate a divisor to a differential form. Let $P \in X$ and U be a coordinate neighborhood around P. Let z denote the local coordinate on U. The differential form on U is of the form $f(z)dz$. Define the order of ω at P by $\mathrm{ord}_P(\omega) = \mathrm{ord}_P(f)$. Define:

$$\mathrm{div}(\omega) = \sum_P \mathrm{ord}_P(\omega) \cdot P.$$

Notice that $\mathrm{div}(f\omega) = \mathrm{div}(f) + \mathrm{div}(\omega)$.

Let $\mathcal{M}(X)$ denote the set of meromorphic functions on X. For any divisor D on X, we define the set

$$L(D) = \{f \in \mathcal{M}(X) \mid \mathrm{div}(f) + D \geq 0\} \cup \{0\}$$

It is easy to see that this is a vector space. In fact $L(D)$ is finite dimensional. Its dimension is denoted by $l(D)$.

Definition 6.9 (Canonical divisor)**.** For any compact Riemann surface X, a canonical divisor is a divisor $\mathrm{div}(\omega)$ where ω is a one form on X.

The Riemann Roch theorem helps us to calculate the dimension of the space $L(D)$ for any divisor D. In this way, given a specific number of zeroes and poles, we can define a divisor D on X with appropriate coefficients n_i, so that the space $L(D)$ contains the functions with the given number of zeroes and poles. The Riemann Roch theorem therefore helps us calculate the dimension of the space of such functions. The following is the statement of the theorem:

Theorem 6.10 (Riemann-Roch)**.** *Let X be a compact Riemann surface. Let g denote the genus of the surface. Then for any divisor D on X and K a canonical divisor on X, we have that*

$$l(D) = deg(D) + 1 - g + l(K - D)$$

6.1. **Consequences of the Riemann Roch theorem.** In this section we prove three immediate corollaries of the Riemann Roch theorem which will help us prove the dimension formulae.

Corollary 6.11. *A canonical divisor K has degree $2g - 2$ and $l(K) = g$.*

Proof. If we take $D = 0$, the space $L(D)$ is the space of holomorphic functions. By fact 6.1, we have that $L(D)$ consists of constant functions. Thus $l(0) = 1$. Substituting everything in theorem 6.10 we get that $l(K) = g$.

When we put $D = K$, then $l(K) = \deg(K) + 1 - g + l(0)$. Putting $l(K) = 1$, we have that $\deg(K) = 2g - 2$. □

Corollary 6.12. *The degree of a k-fold differential form is $k(2g - 2)$.*

Proof. Suppose that $\lambda \in \Omega^1(X(\Gamma))$ is a non zero 1-form. We know that its degree is $2g-2$. Notice that $\lambda^k \in \Omega^k(X(\Gamma))$ and its divisor has degree $k(2g-2)$. This implies that $\Omega^k(X(\Gamma)) = C(X(\Gamma))\lambda^k$. Since for all $f \in C(X(\Gamma))$, $\deg(\mathrm{div}(f)) = 0$, it follows that every non zero differential form $\omega \in \Omega^k(X(\Gamma))$ has degree $k(2g - 2)$. □

Corollary 6.13. *If $deg(D) > 2g - 2$, then $l(D) = deg(D) + 1 - g$*

Proof. If $\deg(D) > 2g-2$, then $\deg(K-D) < 0$. This implies that for any non zero $f \in \mathcal{M}(X)$, $\deg(f+K-D) < 0$. Therefore, $\mathrm{div}(f+K-D)$ will never be a positive divisor. Thus, $L(K-D) = 0$ so that $l(K-D) = 0$ and the result follows. □

Corollary 6.14. *If $deg(D) < 0$, then $l(D) = 0$.*

Proof. The proof of this corollary is along similar lines to the proof of corollary 6.13. □

6.2. **Order of a modular form and its corresponding differential form.** We know that the local homeomorphism around the elliptic points "looks like" $z \mapsto z^e$, similarly for the cusps the local map looks like $z \mapsto e^{2\pi iz/h}$. Therefore when we look at the corresponding differential form, the order of the differential form at the cusps and the elliptic points will not be the same as the order of the modular form at those points. We calculate the order keeping in mind the Riemann structure as follows:

Lemma 6.15. *Let $f \in \mathcal{A}_{2k}(\Gamma)$ and let ω be the corresponding k-fold differential form on $X(\Gamma)$. Let Π be the projection map as defined previously. Suppose $\tau \in \hat{\mathbb{H}}$.*

(1) If τ is an elliptic point with period e, then

$$ord_\tau(f) = ord_{\Pi(\tau)}(\omega)e + k(e - 1)$$

(2) If τ is a cusp point with width h, then

$$ord_\tau(f) = ord_{\Pi(\tau)}(\omega) + k$$

(3) If τ is any other point then

$$ord_\tau(f) = ord_{\Pi(\tau)}(\omega)$$

Proof of 1. Let U be a coordinate neighborhood of τ and $\Pi(U)$ be the neighborhood around $\Pi(\tau)$. We then have the following commutative diagram:

$$\begin{tikzcd} U \arrow[r, "\delta"] \arrow[d, "\Pi"'] & U' \arrow[d, "\rho"] \\ \Pi(U) \arrow[r, "\sim"'] & V \end{tikzcd}$$

The transformation δ is defined as in section 4.4 and the the map ρ is given by $\rho(z) = z^e$. Since $\Pi(U)$ is homeomorphic to V, the k-fold differential form ω can be considered on V. Let $\omega = g(z)dz^k$ for some meromorphic function g on V. Thus $\text{ord}(\omega) = \text{ord}(g)$ at the points in V. We have that

$$\rho^*(\omega) = g(\rho(z))d(\rho(z))^k = g(z^e)d(z^e)^k = h^k g(z^e)z^{k(e-1)}(dz)^k$$

This further implies that

$$\text{ord}_\tau(\rho^*(\omega)) = \text{ord}_\tau(f) = \text{ord}_{\Pi(\tau)}(\omega)e + k(e-1).$$

□

Proof of 2. Let U be a local coordinate around a cusp. The following commutative diagram shows the local map.

$$\begin{tikzcd} U \arrow[r, "\delta"] \arrow[d, "\Pi"'] & U' \arrow[d, "q"] \\ \Pi(U) \arrow[r, "\sim"'] & V \end{tikzcd}$$

In the above diagram, δ is defined as in section 4.7 and the map q is such that $q(\tau) = e^{2\pi i\tau/h}$. Suppose $\omega = g(q)(dq)^k$ be a k-fold differential form on V. Then $dq = (2\pi i/h)qdz$. This implies that the pullback of ω on $\hat{\mathbb{H}}$ is $(2\pi i/h)g(q(z))q(z)^k(dz)^k$. Thus ω corresponds to the modular form $f(z) = g(q(z))q(z)^k$. This further implies that $\text{ord}_\tau(f) = \text{ord}_{\Pi(\tau)}(\omega) + k$.

□

Proof of 3. In this case, Π is a local homeomorphism. We therefore have that $\text{ord}_{\Pi(\tau)}(\omega) = \text{ord}_\tau(f)$. □

Keeping the above formulae in mind, we now finally move on to applying the Riemann Roch theorem to find the dimension of the space of modular forms.

7. The dimension formula

All throughout this section, f is a modular form of weight $2k$. The aim of this section is to prove the following theorem:

Theorem 7.1. *Suppose the modular curve $X(\Gamma)$ has a genus g for some congruence subgroup Γ. Let v_∞ denote the number of cusps of $X(\Gamma)$. Let ϵ_2 and ϵ_3 be the number of elliptic points of period 2 and 3 respectively on $X(\Gamma)$. Then the dimension of $\mathcal{M}_{2k}(\Gamma)$ is given by:*

$$dim(\mathcal{M}_{2k}(\Gamma)) = \begin{cases} 0 & \text{if } k < 0 \\ 1 & \text{if } k = 0 \\ (2k-1)(g-1) + v_\infty k + \lfloor k/2 \rfloor \epsilon_2 + \lfloor 2k/3 \rfloor \epsilon_3 & \text{if } k > 0 \end{cases}$$

Proof. We will first characterize the image of the subspace $\mathcal{M}_{2k}(\Gamma) \subset \mathcal{A}_{2k}(\Gamma)$ under the isomorphism theorem 5.3. Suppose $f \in \mathcal{M}_{2k}(\Gamma)$. Let ω be its corresponding k-fold differential form on $X(\Gamma)$. Since f is holomorphic, $\mathrm{ord}_\tau(f) \geq 0$ for all $\tau \in \hat{\mathbb{H}}$. Let ω_0 be a nonzero k-fold differential form so that $\omega = h\omega_0$ for some function $h \in C(X(\Gamma))$. Thus $\mathrm{ord}_{\Pi(\tau)}(\omega) = \mathrm{ord}_{\Pi(\tau)}(h\omega_0)$. From the previous section relating the order of zeroes and poles of ω with f we have that:

$$\text{When } \tau \text{ is an elliptic point: } \mathrm{ord}_{\Pi(\tau)}(h) + \mathrm{ord}_{\Pi(\tau)}(\omega_0) + k(1 - 1/e) \geq 0$$

$$\text{When } \tau \text{ is a cusp: } \mathrm{ord}_{\Pi(\tau)}(h) + \mathrm{ord}_{\Pi(\tau)}(\omega_0) + k \geq 0$$

$$\text{Otherwise: } \mathrm{ord}_{\Pi(\tau)}(h) + \mathrm{ord}_{\Pi(\tau)}(\omega_0) \geq 0$$

The above three equations characterize the subspace $\mathcal{M}_{2k}(\Gamma)$. Combining the three equations and rewriting them in the language of divisors we arrive at:

$$\mathrm{div}(h) + \mathrm{div}(\omega_0) + \sum_{\tau \text{ cusp}} k \cdot \tau + \sum_{\tau \text{ elliptic}} \lfloor k(1 - 1/e_i) \rfloor \cdot \tau_i \geq 0$$

The symbol e_i refers to the period of the elliptic point τ_i. Let D be the divisor given by:

$$D = \mathrm{div}(\omega_0) + \sum_{\tau \text{ cusp}} k \cdot \tau + \sum_{\tau \text{ elliptic}} \lfloor k(1 - 1/e_i) \rfloor \cdot \tau_i$$

Then the above condition characterizing the space of holomorphic modular forms reduces to the following:

$$\mathrm{div}(h) + D \geq 0$$

In other words, because of the isomorphism of Theorem 5.3 we have a one to one correspondence between the sets:

$$\{f \mid f \in \mathcal{M}_{2k}(\Gamma)\} \leftrightarrow \{h \in C(X(\Gamma)) \mid \mathrm{div}(h + D) \geq 0\}.$$

We therefore deduce that $L(D) \cong \mathcal{M}_{2k}(\Gamma)$. We are now ready to apply the Riemann Roch theorem to find the dimension $l(D)$ which will give us the dimension for the space $\mathcal{M}_{2k}(\Gamma)$. Since ω_0 is a k-fold differential form, its degree is $k(2g - 2)$ by corollary 6.12. This implies that:

$$\deg(D) = k(2g - 2) + v_\infty k + \lfloor k/2 \rfloor \epsilon_2 + \lfloor 2k/3 \rfloor \epsilon_3.$$

Notice that when $k > 0, \deg(D) > (2g-2)$. This implies that $l(D) = \deg(D) - g + 1$ by corollary 6.13. When $k > 0$ we finally get that:

$$\dim(\mathcal{M}_{2k}(\Gamma)) = l(D) = (2k - 1)(g - 1) + v_\infty k + \lfloor k/2 \rfloor \epsilon_2 + \lfloor 2k/3 \rfloor \epsilon_3.$$

When $k < 0, \deg(D) < 0$. By corollary 6.14 $l(D) = 0$. In the case when $k = 0$, modular forms are well defined holomorphic functions on the modular curve. By fact 6.1, they are constant functions. Therefore $\dim(\mathcal{M}_0(\Gamma)) = 1$. □

We saw that the dimension can be calculated if we know the number of cusps, elliptic points and the genus of the associated modular curve. Using sage we compute the dimension of the space of modular forms for some subgroups:

```
G=SL2Z
c=G.ncusps()
g=G.genus()
e2=G.nu2()
e3=G.nu3()
var('k')
```

```
f(k)=(2*k-1)*(g-1)+(k*c)+floor((k/2)*e2)+floor(((2*k)/3)*e3)
f(5)
OUTPUT:1
```

The above code calculates the dimension of the space $\mathcal{M}_{10}(SL_2(\mathbb{Z}))$. We saw that the Eisenstein series $G_{10}(\tau)$ is a modular form of weight 10. Since the space is one dimensional, every element in $\mathcal{M}_{10}(SL_2(\mathbb{Z}))$ is a multiple of $G_{10}(\tau)$. Notice that the product $G_4(\tau)G_6(\tau)$ is a weight 10 modular form. We can therefore write $G_4(\tau)G_6(\tau)$ as a multiple of $G_{10}(\tau)$. The relations arising between Eisenstein series as a result of the dimension formula help us to prove many combinatorial identities.

We can similarly calculate the dimension of the space $\mathcal{M}_k(\Gamma(N))$. For example the dimension of the space $\mathcal{M}_{10}(\Gamma(13))$ is calculated to be 861.

```
n=13
G=Gamma(n)
c=Gamma(n).ncusps()
g=Gamma(n).genus()
e2=Gamma(n).nu2()
e3=Gamma(n).nu3()
var('k')
```

```
f(k)=(2*k-1)*(g-1)+(k*c)+floor((k/2)*e2)+floor(((2*k)/3)*e3)
f(5);
OUTPUT:861
```

Sage computes the dimension directly as well.

```
ModularForms(Gamma1(17),10);
```

The above command helps sage to create the space of modular forms and the output is:

```
Modular Forms space of dimension 116 for Congruence Subgroup
Gamma1(17) of weight 10 over Rational Field
```

Acknowledgments

It is a pleasure to thank my mentor Drew Moore for patiently attending to all my questions and helping me throughout this program. I would also like to thank Dr. Abhik Ganguli for introducing me to the beautiful mathematics of modular forms. Above all, to Professor May, for continuing to inspire us through the program. This REU would not have been possible without you. Thank you.

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